

Project Design Phase-II Technology Stack (Architecture & Stack)

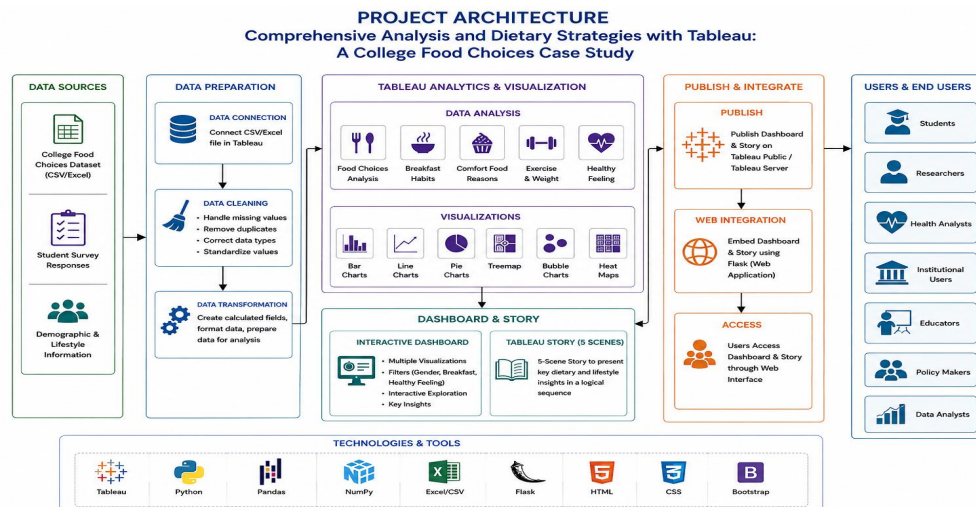
Date	04 July 2026
Team ID	
Project Name	Comprehensive Analysis Dietary Strategies with Tableau:A college food choices case study
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Order processing during pandemics for offline mode

Reference: <https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/>



Guidelines:

- Follow a structured process flow: **Collection, Preparation, Analysis & Visualization.**
- Ensure seamless integration of **Tableau with the dataset** for interactive insights.
- Include **calculated fields** to derive meaningful metrics for analysis.
- Create an **interactive dashboard** with multiple visualizations and filters for user exploration.
- Develop a **7-scene Tableau Story** to present major findings in a logical sequence.

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Web interface for users to access the College Food Choices dashboard and Tableau Story.	HTML5, CSS3, Bootstrap
2.	Application Logic-1	Handles web integration and embeds the Tableau dashboard and Story in the web application.	Python, Flask
3.	Application Logic-2	Performs data collection, preparation, cleaning, and transformation for analysis.	Python, Pandas, NumPy
4.	Application Logic-3	Performs data analysis, visualization, dashboard creation, and Story development.	Tableau
5.	Datasource	Stores the College Food Choices dataset containing student dietary, demographic, lifestyle, and health-related information.	CSV / Excel
6.	Data Visualization	Creates interactive charts and visualizations to analyze food choices and lifestyle patterns.	Tableau

7.	Interactive Dashboard	Displays key dietary and lifestyle insights with interactive filters and multiple visualizations.	Tableau Dashboard
8.	Tableau Story	Presents major project findings through a structured 5-scene Story.	Tableau Story
9.	Web Integration	Publishes and embeds the Tableau dashboard and Story for user access	Tableau Public / Flask
10.	Data Analytics	Analyzes breakfast habits, food choices, comfort food, exercise, weight, and healthy-feeling patterns.	Tableau, Calculated Fields
11.	Infrastructure / Hosting	Provides an environment for running the web application and accessing the published analytics.	Flask Web Server, Tableau Public

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Open-source tools are used for data processing and web integration.	Python, Pandas, NumPy, Flask, Bootstrap
2.	Data Accuracy	Data is validated, cleaned, and prepared before analysis to ensure reliable visualizations and insights.	Tableau, Pandas
3.	Interactive Architecture	The solution connects the dataset, data preparation, Tableau analytics, dashboard, Story, and web interface in a structured workflow.	Tableau, Flask, HTML, CSS
4.	Availability	The published dashboard and Story can be accessed by intended users through the web interface.	Tableau Public, Flask
5.	Performance	The solution is designed for efficient data rendering, visualization loading, and interactive filtering.	Tableau, Pandas, NumPy

S.No	Characteristics	Description	Technology
6.	Usability	The dashboard provides interactive filters and clear visualizations for easy exploration of dietary insights.	Tableau Dashboard
7.	Usability	The solution can be extended with larger datasets, additional colleges, and more dietary or lifestyle variables.	Tableau, Python, Flask

References:

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>